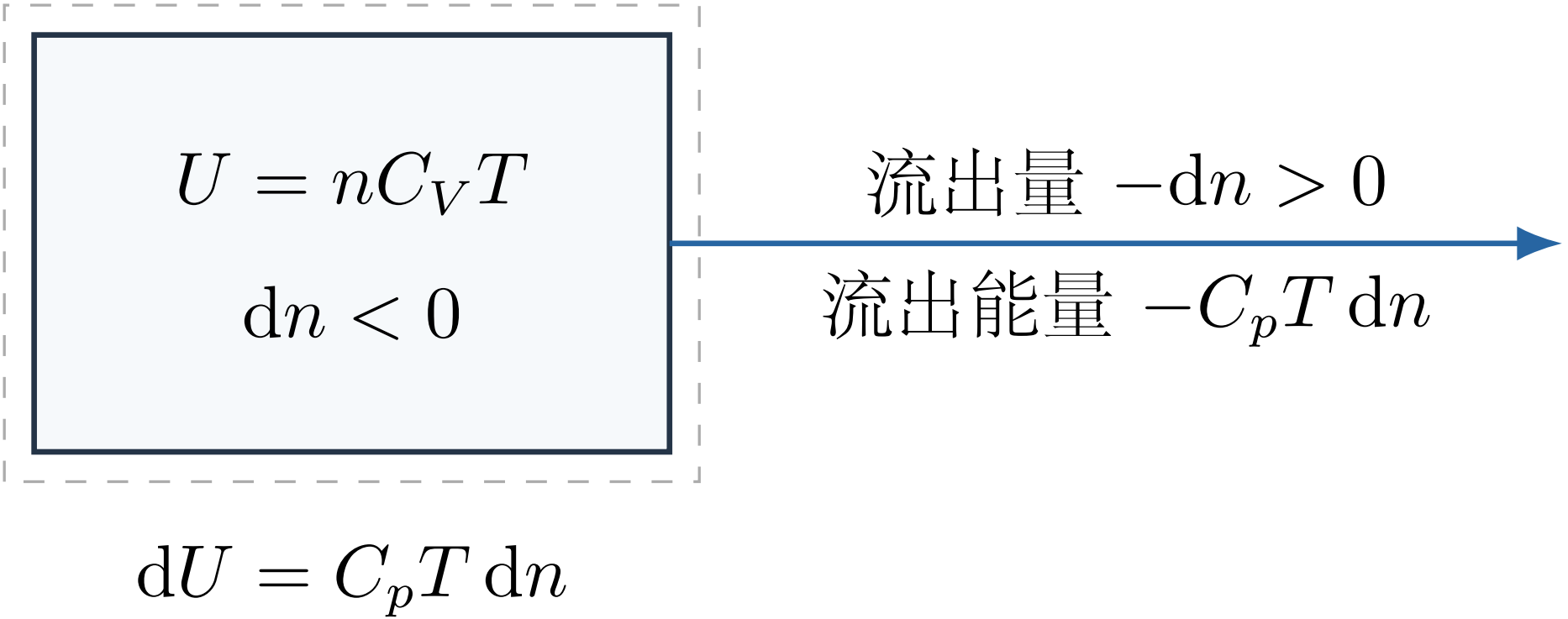

$$\mathrm{d}U = C_pT\,\mathrm{d}n$$